

VALIDATION PLAN

OPERATIONAL QUALIFICATION

ACCEPTANCE SAMPLING MODULE

JR Validated Environment — Acceptance Sampling Module v1.0

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Version History

Version	Date	Author	Description
1.0	2026-03-19	Joep Rous	Initial release. Covers all 4 AS scripts; 44 automated test cases.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-19
Reviewer	[Name]		
Approver (QA)	[Name]		

Purpose and Scope

This Operational Qualification (OQ) Validation Plan defines the test cases for the Acceptance Sampling (AS) module of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability needed to demonstrate that each AS script performs correctly for its intended use under 21 CFR 820.70(i) and ISO 13485.

This plan covers 44 automated test cases executed using the pytest framework via the admin_as_oq runner. Test evidence is written to ~/jrscript/<PROJECT_ID>/validation/as_oq_execution_<timestamp>.txt.

User Requirements

The following user requirements are tested by this plan.

UR ID	Requirement
UR-AS-001	jr_as_attributes shall accept valid lot size, AQL, and RQL arguments and exit with code 0.
UR-AS-002	jr_as_attributes shall produce a single sampling plan (n and c) in the output.
UR-AS-003	jr_as_attributes shall produce a double sampling plan (n1, c1, c2) in the output when a valid plan exists.
UR-AS-004	jr_as_attributes shall produce an OC curve table with p and Pa values.
UR-AS-005	jr_as_attributes shall save a PNG plot to ~/Downloads/.
UR-AS-006	jr_as_attributes shall exit non-zero and print usage when called with no arguments.
UR-AS-007	jr_as_attributes shall exit non-zero when aql ≥ rql.
UR-AS-008	jr_as_attributes shall exit non-zero when aql > 1 or < 0.
UR-AS-009	jr_as_attributes shall exit non-zero when lot_size < 2.
UR-AS-010	jr_as_attributes shall exit non-zero when --alpha is out of range.
UR-AS-011	jr_as_attributes shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-AS-012	jr_as_variables shall accept valid lot size, AQL, RQL, and --sides arguments and exit with code 0.
UR-AS-013	jr_as_variables shall produce an acceptability constant k in the output and it shall be positive.
UR-AS-014	jr_as_variables shall produce a valid plan for --sides 2 with positive k.
UR-AS-015	jr_as_variables shall produce an OC curve table with Pa values.
UR-AS-016	jr_as_variables shall save a PNG plot to ~/Downloads/.
UR-AS-017	jr_as_variables shall exit non-zero when called with no arguments.
UR-AS-018	jr_as_variables shall exit non-zero when aql ≥ rql.
UR-AS-019	jr_as_variables shall exit non-zero when --sides value is not 1 or 2.
UR-AS-020	jr_as_variables shall produce output indicating a sample size reduction compared to the equivalent attributes plan.
UR-AS-021	jr_as_variables shall exit non-zero when --alpha is out of range.
UR-AS-022	jr_as_variables shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-AS-023	jr_as_oc_curve shall accept valid n and c arguments and exit with code 0 with Pa values in output.
UR-AS-024	jr_as_oc_curve shall produce Pa > 0.99 at p = 0.001 for n=32, c=1.
UR-AS-025	jr_as_oc_curve shall produce Pa values that decrease monotonically as p increases.
UR-AS-026	jr_as_oc_curve shall accept the --lot-size flag without error.
UR-AS-027	jr_as_oc_curve shall save a PNG plot to ~/Downloads/.
UR-AS-028	jr_as_oc_curve shall accept --aql and --rql flags and display AQL in output.
UR-AS-029	jr_as_oc_curve shall exit non-zero when called with no arguments.
UR-AS-030	jr_as_oc_curve shall exit non-zero when c ≥ n.
UR-AS-031	jr_as_oc_curve shall exit non-zero when n ≤ 0.
UR-AS-032	jr_as_oc_curve shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.

UR-AS-033	jrc_as_evaluate attributes mode shall produce ACCEPT verdict for a lot with defective count $\leq c$.
UR-AS-034	jrc_as_evaluate attributes mode shall produce REJECT verdict for a lot with defective count $> c$.
UR-AS-035	jrc_as_evaluate variables mode shall produce ACCEPT verdict for a lot with Q-statistic $\geq k$.
UR-AS-036	jrc_as_evaluate variables mode shall produce REJECT verdict for a lot with Q-statistic $< k$.
UR-AS-037	jrc_as_evaluate variables mode shall save a PNG plot to ~/Downloads/.
UR-AS-038	jrc_as_evaluate attributes mode shall save a PNG plot to ~/Downloads/.
UR-AS-039	jrc_as_evaluate shall exit non-zero when --type is missing.
UR-AS-040	jrc_as_evaluate attributes mode shall exit non-zero when --c is missing.
UR-AS-041	jrc_as_evaluate variables mode shall exit non-zero when --k is missing.
UR-AS-042	jrc_as_evaluate attributes mode shall exit non-zero and report the missing column when the data file lacks a 'result' column.
UR-AS-043	jrc_as_evaluate variables mode shall exit non-zero and report the missing column when the data file lacks a 'value' column.
UR-AS-044	jrc_as_evaluate shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.

Test Environment

All test cases are executed via the admin_as_oq runner using pytest with the shared OQ virtual environment at ~/venvs/<PROJECT_ID>_oq/. Evidence is written to ~/.jrscrip/<PROJECT_ID>/validation/.

Item	Specification
Language	R ($\geq 4.3.0$ recommended)
Test framework	pytest (≥ 8.0)
Test runner	admin_as_oq
OQ venv	~/venvs/<PROJECT_ID>_oq/ (shared with admin_oq)
Evidence output	~/.jrscrip/<PROJECT_ID>/validation/as_oq_execution_<timestamp>.txt
Validation plan ID	JR-VP-AS-001

Test Cases — jrc_as_attributes

TC-AS-ATTR-001 Happy path — exit 0

Tests UR(s)	UR-AS-001
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Attributes Sampling Plan' or equivalent plan header.

TC-AS-ATTR-002 Single plan n and c present

Tests UR(s)	UR-AS-002
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the single sampling plan outputs both sample size n and acceptance number c.
Pass criterion	Output contains 'Sample size' with a positive integer, and 'Acceptance number' with a nonnegative integer.

TC-AS-ATTR-003 Double sampling plan present

Tests UR(s)	UR-AS-003
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the double sampling plan section is present in the output.
Pass criterion	Output contains 'Double Sampling Plan' or 'Stage 1'.

TC-AS-ATTR-004 OC curve table present

Tests UR(s)	UR-AS-004
Command / action	<code>jrc_as_attributes 500 0.01 0.10</code>
Rationale	Confirms the OC curve table is included in the output.
Pass criterion	Output contains 'OC Curve' and 'Pa' header.

TC-AS-ATTR-005 PNG saved to ~/Downloads/

Tests UR(s)	UR-AS-005
Command / action	<code>jrc_as_attributes 500 0.01 0.10</code>
Rationale	Confirms a PNG file is created in ~/Downloads/ following the script run.
Pass criterion	A file matching *_jrc_as_attributes.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-ATTR-006 No arguments — non-zero exit

Tests UR(s)	UR-AS-006
Command / action	<code>jrc_as_attributes</code> (no arguments)
Rationale	Confirms the script exits non-zero and displays usage guidance when called with no arguments.
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-007 aql ≥ rql — non-zero exit

Tests UR(s)	UR-AS-007
Command / action	<code>jrc_as_attributes 500 0.10 0.01</code>
Rationale	Confirms the script detects and rejects inverted AQL/RQL inputs.
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-008 aql > 1 — non-zero exit

Tests UR(s)	UR-AS-008
Command / action	<code>jrc_as_attributes 500 1.5 0.10</code>
Rationale	Confirms the script rejects an AQL value out of the (0, 1) range.
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-009 lot_size = 1 — non-zero exit

Tests UR(s)	UR-AS-009
Command / action	<code>jrc_as_attributes 1 0.01 0.10</code>
Rationale	Confirms the script rejects a lot size below the minimum (2).
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-010 --alpha out of range — non-zero exit

Tests UR(s)	UR-AS-010
Command / action	<code>jrc_as_attributes 500 0.01 0.10 --alpha 2.0</code>
Rationale	Confirms the script rejects an --alpha value outside (0, 1).
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-011 Bypass protection

Tests UR(s)	UR-AS-011
Command / action	<code>Rscript repos/as/R/jrc_as_attributes.R 500 0.01 0.10</code> (<code>RENV_PATHS_ROOT</code> unset)
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions <code>RENV_PATHS_ROOT</code> in the error output.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

Test Cases — jrc_as_variables

TC-AS-VAR-001 Happy path — exit 0, Variables Plan in output

Tests UR(s)	UR-AS-012
Command / action	<code>jrc_as_variables 500 0.01 0.10</code>
Rationale	Confirms valid one-sided inputs produce exit 0 and a plan header.

Pass criterion	Exit code = 0. Output contains 'Variables Plan' or 'Variables Sampling Plan'.
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TC-AS-VAR-002 k value present and positive

Tests UR(s)	UR-AS-013
Command / action	<code>jrc_as_variables 500 0.01 0.10</code>
Rationale	Confirms the acceptability constant k is reported and is positive.
Pass criterion	Output contains 'constant (k)' or 'k' or 'k ='. If a numeric k is found, $k > 0$.

TC-AS-VAR-003 --sides 2 produces valid plan

Tests UR(s)	UR-AS-014
Command / action	<code>jrc_as_variables 500 0.01 0.10 --sides 2</code>
Rationale	Confirms two-sided specification produces a valid plan with exit 0 and positive k.
Pass criterion	Exit code = 0. Output contains 'Sides: 2'. If k extracted, $k > 0$.

TC-AS-VAR-004 OC curve table present

Tests UR(s)	UR-AS-015
Command / action	<code>jrc_as_variables 500 0.01 0.10</code>
Rationale	Confirms the OC curve table is present in the output.
Pass criterion	Output contains 'OC Curve' and 'Pa'.

TC-AS-VAR-005 PNG saved to ~/Downloads/

Tests UR(s)	UR-AS-016
Command / action	<code>jrc_as_variables 500 0.01 0.10</code>
Rationale	Confirms a PNG file is created in ~/Downloads/ following the script run.
Pass criterion	A file matching *_jrc_as_variables.png exists in ~/Downloads/ with mtime $\geq$ test start time.

TC-AS-VAR-006 No arguments — non-zero exit

Tests UR(s)	UR-AS-017
Command / action	<code>jrc_as_variables (no arguments)</code>
Rationale	Confirms the script exits non-zero when called with no arguments.
Pass criterion	Exit code $\neq 0$.

TC-AS-VAR-007 aql $\geq$ rql — non-zero exit

Tests UR(s)	UR-AS-018
Command / action	<code>jrc_as_variables 500 0.10 0.01</code>
Rationale	Confirms the script rejects inverted AQL/RQL.
Pass criterion	Exit code $\neq 0$.

TC-AS-VAR-008 --sides invalid — non-zero exit

Tests UR(s)	UR-AS-019
Command / action	<code>jrc_as_variables 500 0.01 0.10 --sides 3</code>
Rationale	Confirms the script rejects --sides values other than 1 or 2.
Pass criterion	Exit code $\neq 0$.

TC-AS-VAR-009 Sample reduction noted in output

Tests UR(s)	UR-AS-020
Command / action	<code>jrc_as_variables 500 0.01 0.10</code>
Rationale	Confirms the output includes a comparison with the equivalent attributes plan.
Pass criterion	Output contains 'Sample reduction', 'fewer', or 'Comparison'.

TC-AS-VAR-010 --alpha out of range — non-zero exit

Tests UR(s)	UR-AS-021
Command / action	<code>jrc_as_variables 500 0.01 0.10 --alpha 2.0</code>
Rationale	Confirms the script rejects an --alpha value outside (0, 1).
Pass criterion	Exit code $\neq 0$.

TC-AS-VAR-011 Bypass protection

Tests UR(s)	UR-AS-022
Command / action	Rscript repos/as/R/jrc_as_variables.R 500 0.01 0.10 (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code $\neq$ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

Test Cases — jrc_as_oc_curve

TC-AS-OCC-001 Happy path — exit 0, Pa values in output

Tests UR(s)	UR-AS-023
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms valid n and c produce exit 0 and Pa values.
Pass criterion	Exit code = 0. Output contains 'Pa' and at least one numeric Pa value.

TC-AS-OCC-002 Pa at p=0.001 is > 0.99

Tests UR(s)	UR-AS-024
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms the plan (n=32, c=1) has very high acceptance probability at very low defect rates.
Pass criterion	Pa value in the first data row (p=0.001) extracted from output is > 0.99.

TC-AS-OCC-003 Pa decreases as p increases

Tests UR(s)	UR-AS-025
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms the OC curve is monotonically decreasing in p.
Pass criterion	Pa value at first p in the table > Pa value at last p in the table.

TC-AS-OCC-004 --lot-size accepted

Tests UR(s)	UR-AS-026
Command / action	jrc_as_oc_curve 32 1 --lot-size 200
Rationale	Confirms the --lot-size flag is accepted without error.
Pass criterion	Exit code = 0.

TC-AS-OCC-005 PNG saved to ~/Downloads/

Tests UR(s)	UR-AS-027
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms a PNG file is created in ~/Downloads/ following the script run.
Pass criterion	A file matching *_jrc_as_oc_curve.png exists in ~/Downloads/ with mtime $\geq$ test start time.

TC-AS-OCC-006 --aql and --rql accepted, AQL in output

Tests UR(s)	UR-AS-028
Command / action	jrc_as_oc_curve 32 1 --aql 0.01 --rql 0.10
Rationale	Confirms --aql and --rql flags are accepted and reflected in output.
Pass criterion	Exit code = 0. Output contains 'AQL'.

TC-AS-OCC-007 No arguments — non-zero exit

Tests UR(s)	UR-AS-029
Command / action	jrc_as_oc_curve (no arguments)
Rationale	Confirms the script exits non-zero when called with no arguments.
Pass criterion	Exit code $\neq$ 0.

TC-AS-OCC-008 c $\geq$ n — non-zero exit

Tests UR(s)	UR-AS-030
Command / action	jrc_as_oc_curve 10 10
Rationale	Confirms the script rejects c $\geq$ n (acceptance number must be less than sample size).

Pass criterion	Exit code $\neq$ 0.
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TC-AS-OCC-009 $n \leq 0$ — non-zero exit

Tests UR(s)	UR-AS-031
Command / action	<code>jrc_as_oc_curve 0 0</code>
Rationale	Confirms the script rejects $n \leq 0$.
Pass criterion	Exit code $\neq$ 0.

TC-AS-OCC-010 Bypass protection

Tests UR(s)	UR-AS-032
Command / action	<code>Rscript repos/as/R/jrc_as_oc_curve.R 32 1 (RENV_PATHS_ROOT unset)</code>
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code $\neq$ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

Test Cases — jrc_as_evaluate

TC-AS-EVAL-001 Attributes ACCEPT verdict

Tests UR(s)	UR-AS-033
Command / action	<code>jrc_as_evaluate oq/data/attr_accept_lot.csv --type attributes --c 2</code>
Rationale	Confirms ACCEPT verdict for a lot with 1 defective against $c=2$. The test lot (attr_accept_lot.csv) contains 32 units with 1 defective.
Pass criterion	Exit code = 0. Output contains 'ACCEPT'.

TC-AS-EVAL-002 Attributes REJECT verdict

Tests UR(s)	UR-AS-034
Command / action	<code>jrc_as_evaluate oq/data/attr_reject_lot.csv --type attributes --c 2</code>
Rationale	Confirms REJECT verdict for a lot with 5 defectives against $c=2$. The test lot (attr_reject_lot.csv) contains 32 units with 5 defectives.
Pass criterion	Exit code = 0. Output contains 'REJECT'.

TC-AS-EVAL-003 Variables ACCEPT verdict

Tests UR(s)	UR-AS-035
Command / action	<code>jrc_as_evaluate oq/data/var_accept_lot.csv --type variables --k 1.45 --lsl 9.5</code>
Rationale	Confirms ACCEPT verdict for a lot where $Q_L \gg k$. The test lot (var_accept_lot.csv) has mean ≈ 10.45 and SD ≈ 0.051 , giving $Q_L \approx 18.5$.
Pass criterion	Exit code = 0. Output contains 'ACCEPT'.

TC-AS-EVAL-004 Variables REJECT verdict

Tests UR(s)	UR-AS-036
Command / action	<code>jrc_as_evaluate oq/data/var_reject_lot.csv --type variables --k 1.45 --lsl 9.5</code>
Rationale	Confirms REJECT verdict for a lot where $Q_L < k$. The test lot (var_reject_lot.csv) has mean ≈ 9.6 and SD ≈ 0.103 , giving $Q_L \approx 0.975$.
Pass criterion	Exit code = 0. Output contains 'REJECT'.

TC-AS-EVAL-005 Variables mode PNG saved

Tests UR(s)	UR-AS-037
Command / action	<code>jrc_as_evaluate oq/data/var_accept_lot.csv --type variables --k 1.45 --lsl 9.5</code>
Rationale	Confirms a PNG is created in <code>~/Downloads/</code> after a variables mode run.
Pass criterion	A file matching <code>*_jrc_as_evaluate.png</code> exists in <code>~/Downloads/</code> with mtime $\geq$ test start time.

TC-AS-EVAL-006 Attributes mode PNG saved

Tests UR(s)	UR-AS-038
Command / action	<code>jrc_as_evaluate oq/data/attr_accept_lot.csv --type attributes --c 2</code>
Rationale	Confirms a PNG is created in ~/Downloads/ after an attributes mode run.
Pass criterion	A file matching *_jrc_as_evaluate.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-EVAL-007 Missing --type — non-zero exit

Tests UR(s)	UR-AS-039
Command / action	<code>jrc_as_evaluate oq/data/attr_accept_lot.csv --c 2</code>
Rationale	Confirms the script exits non-zero when --type is omitted.
Pass criterion	Exit code ≠ 0.

TC-AS-EVAL-008 Attributes mode missing --c — non-zero exit

Tests UR(s)	UR-AS-040
Command / action	<code>jrc_as_evaluate oq/data/attr_accept_lot.csv --type attributes</code>
Rationale	Confirms the script exits non-zero when --type attributes is specified without --c.
Pass criterion	Exit code ≠ 0.

TC-AS-EVAL-009 Variables mode missing --k — non-zero exit

Tests UR(s)	UR-AS-041
Command / action	<code>jrc_as_evaluate oq/data/var_accept_lot.csv --type variables --lsl 9.5</code>
Rationale	Confirms the script exits non-zero when --type variables is specified without --k.
Pass criterion	Exit code ≠ 0.

TC-AS-EVAL-010 Attributes: missing 'result' column

Tests UR(s)	UR-AS-042
Command / action	<code>jrc_as_evaluate oq/data/attr_missing_result.csv --type attributes --c 2</code>
Rationale	Confirms the script exits non-zero and reports the missing 'result' column when an attributes CSV has the wrong column name.
Pass criterion	Exit code ≠ 0. 'result' appears in combined stdout/stderr.

TC-AS-EVAL-011 Variables: missing 'value' column

Tests UR(s)	UR-AS-043
Command / action	<code>jrc_as_evaluate oq/data/var_missing_value.csv --type variables --k 1.45 --lsl 9.5</code>
Rationale	Confirms the script exits non-zero and reports the missing 'value' column when a variables CSV has the wrong column name.
Pass criterion	Exit code ≠ 0. 'value' appears in combined stdout/stderr.

TC-AS-EVAL-012 Bypass protection

Tests UR(s)	UR-AS-044
Command / action	<code>Rscript repos/as/R/jrc_as_evaluate.R oq/data/attr_accept_lot.csv ... (RENV_PATHS_ROOT unset)</code>
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

Requirements Traceability Matrix

Maps each test case to the user requirement(s) it covers.

Test Case ID	User Requirement(s)
TC-AS-ATTR-001	UR-AS-001
TC-AS-ATTR-002	UR-AS-002
TC-AS-ATTR-003	UR-AS-003
TC-AS-ATTR-004	UR-AS-004
TC-AS-ATTR-005	UR-AS-005
TC-AS-ATTR-006	UR-AS-006
TC-AS-ATTR-007	UR-AS-007
TC-AS-ATTR-008	UR-AS-008
TC-AS-ATTR-009	UR-AS-009
TC-AS-ATTR-010	UR-AS-010
TC-AS-ATTR-011	UR-AS-011
TC-AS-VAR-001	UR-AS-012
TC-AS-VAR-002	UR-AS-013
TC-AS-VAR-003	UR-AS-014
TC-AS-VAR-004	UR-AS-015
TC-AS-VAR-005	UR-AS-016
TC-AS-VAR-006	UR-AS-017
TC-AS-VAR-007	UR-AS-018
TC-AS-VAR-008	UR-AS-019
TC-AS-VAR-009	UR-AS-020
TC-AS-VAR-010	UR-AS-021
TC-AS-VAR-011	UR-AS-022
TC-AS-OCC-001	UR-AS-023
TC-AS-OCC-002	UR-AS-024
TC-AS-OCC-003	UR-AS-025
TC-AS-OCC-004	UR-AS-026
TC-AS-OCC-005	UR-AS-027
TC-AS-OCC-006	UR-AS-028
TC-AS-OCC-007	UR-AS-029
TC-AS-OCC-008	UR-AS-030
TC-AS-OCC-009	UR-AS-031
TC-AS-OCC-010	UR-AS-032
TC-AS-EVAL-001	UR-AS-033
TC-AS-EVAL-002	UR-AS-034
TC-AS-EVAL-003	UR-AS-035
TC-AS-EVAL-004	UR-AS-036
TC-AS-EVAL-005	UR-AS-037
TC-AS-EVAL-006	UR-AS-038
TC-AS-EVAL-007	UR-AS-039
TC-AS-EVAL-008	UR-AS-040
TC-AS-EVAL-009	UR-AS-041
TC-AS-EVAL-010	UR-AS-042
TC-AS-EVAL-011	UR-AS-043
TC-AS-EVAL-012	UR-AS-044